Exercise: We assume that we will observe n pairs of (x, Y) values, and that the ϵ values are independent. Derive the density for a single Y observation.

$$Y = \beta_0 + \beta_1 x + \sigma\epsilon$$

$$P(Y \le y) = P(\beta_0 + \beta_1 x + \sigma\epsilon \le y)$$

$$= P\left(\epsilon \le \frac{y - (\beta_0 + \beta_1 x)}{\sigma}\right)$$

$$= F_T\left(\frac{y - (\beta_0 + \beta_1 x)}{\sigma}\right), \text{ where } T \sim t_\nu$$

$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y} = f_T\left(\frac{y - (\beta_0 + \beta_1 x)}{\sigma}\right)\left(\frac{1}{\sigma}\right)$$

$$L((\beta_0, \beta_1, \sigma)) = \prod_{i=1}^{n} f_T\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right)\left(\frac{1}{\sigma}\right)$$

In preparation for using `optim()`, here is a function that returns the negative log-likelihood as a function of the three parameters.

```
> # The parameters are beta0, beta1, sigma
>
> negloglikettreg = function(pars,x,y,nu)
+ {
+    resids = y - (x*pars[2] + pars[1])
+    return((-1)*sum(dt(resids/pars[3],nu,log=TRUE))+
+                 length(x)*log(pars[3]))
+ }
```

Note how the `log=TRUE` argument to `dt()` is utilized to directly evaluate the log density.

Exercise: Carefully explain the structure of the above function.

The log likelihood is

$$\log \prod_{i=1}^{n} f_T\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right)\left(\frac{1}{\sigma}\right)$$

$$\log\left(\frac{1}{\sigma}\right) = \log \sigma^{-1} = -\log \sigma$$

$$= \left[\sum_{i=1}^{n} \log f_T\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right)\right] - n \log \sigma$$

= sum(dt(resids/pars[3], nu, log=T))
 − length(x) * log(pars[3])

where resids = y − (x*pars[2] + pars[1])

This function will fit the model using the assumed t-distributed errors.

```
> treg = function(x,y,nu)
+ {
+
+   neglogliketreg = function(pars,x,y,nu)
+   {
+     resids = y - (x*pars[2] + pars[1])
+     return((-1)*sum(dt(resids/pars[3],nu,log=TRUE))+
+             length(x)*log(pars[3]))
+   }
+
+   # Find initial values using least squares
+
+   holdlm = lm(y~x)
```

```
+
+    parsinit = c(holdlm$coef[1],holdlm$coef[2],
+              summary(holdlm)$sigma)
+
+    names(parsinit) = c("beta0","beta1","sigma")
+
+    optout = optim(parsinit,
+              negloglike treg,x=x,y=y,nu=nu,hessian=T)
+
+    list(mle=optout$par, hessian=optout$hessian)
+ }
```


Exercise: Test the t-regression procedure using some simulated examples.

Part 12: Properties of MLEs

Summary

In this Part we explore some of the important properties possessed by maximum likelihood estimators.

Briefly stated, we will see that, asymptotically, these estimators cannot be beat, in the sense that they have minimal MSE, and also are approximatley normally distributed with variance that is easy to calculate. This makes quantifying the error in the estimators straightforward.

Invariance of the MLE

The invariance result can be stated as follows: If $\widehat{\theta}_{\text{MLE}}$ is the MLE for θ, then $g(\widehat{\theta}_{\text{MLE}})$ is the MLE for $g(\theta)$. $= \tau$

This is a useful, important result because it is often the case that we are truly interested in estimating some function of the parameter θ, not θ itself.

Using this, we can say, for example, that if $X_1, X_2, \ldots, X_n$ are i.i.d. normal(μ, σ^2), then the MLE for σ is

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}$$

and the MLE for the signal to noise ratio μ/σ is

$$\overline{X} \Big/ \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}$$

One consequence of this property is that estimation is not affected by how one chooses to parameterize a model.

For example, some choose to represent the Exponential distribution using the "mean parameterization" which leads to $E(X) = \mu$, whereas we have used the "rate parameterization" which leads to $E(X) = 1/\lambda$. Because of invariance, these two approaches will lead to the same estimates, i.e., $\widehat{\mu} = 1/\widehat{\lambda}$.

Exercise: Does unbiasedness have this invariance property?

No. If $\hat{\theta}$ is unbiased for θ, no guarantee that $g(\hat{\theta})$ is an unbiased estimator of $g(\theta)$

e.g. S^2 is unbiased for σ^2, but S is not unbiased for σ

Exercise: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. Exponential(λ). What is the MLE of the tail probability $P(X_i > a)$? Assume $a > 0$.

Let $\tau = P(X_i > a)$. Could rewrite the pdf for X_i in terms of τ, find MLE for τ.

Or use the fact I know MLE for λ is

$\hat{\lambda}_{MLE} = 1/\bar{X}$ and $\tau = P(X_i > a) = e^{-\lambda a}$

So, invariance says $\hat{\tau}_{MLE} = e^{-a/\bar{X}}$

From Casella and Berger, Second Edition

Although estimators such as d_n, called *superefficient*, can be constructed in some generality, they are more of a theoretical oddity than a practical concern. This is because the values of θ for which the variance goes below the bound are a set of Lebesgue measure 0. However, the existence of superefficient estimators serves to remind us to always be careful in our examination of assumptions for establishing properties of estimators (and to be careful in general!).

10.6.2 Suitable Regularity Conditions

The phrase "under suitable regularity conditions" is a somewhat abused phrase, as with enough assumptions we can probably prove whatever we want. However, "regularity conditions" are typically very technical, rather boring, and usually satisfied in most reasonable problems. But they are a necessary evil, so we should deal with them. To be complete, we present a set of regularity conditions that suffice to rigorously establish Theorems 10.1.6 and 10.1.12. These are not the most general conditions but are sufficiently general for many applications (with a notable exception being if the MLE is on the boundary of the parameter space). Be forewarned, the following is not for the fainthearted and can be skipped without sacrificing much in the way of understanding.

These conditions mainly relate to differentiability of the density and the ability to interchange differentiation and integration (as in the conditions for Theorem 7.3.9). For more details and generality, see Stuart, Ord, and Arnold (1999, Chapter 18), Ferguson (1996, Part 4), or Lehmann and Casella (1998, Section 6.3).

The following four assumptions are sufficient to prove Theorem 10.1.6, consistency of MLEs:

($A1$) We observe $X_1, \ldots, X_n$, where $X_i \sim f(x|\theta)$ are iid.

($A2$) The parameter is *identifiable*; that is, if $\theta \neq \theta'$, then $f(x|\theta) \neq f(x|\theta')$.

($A3$) The densities $f(x|\theta)$ have common support, and $f(x|\theta)$ is differentiable in θ.

($A4$) The parameter space Ω contains an open set ω of which the true parameter value θ_0 is an interior point.

The next two assumptions, together with ($A1$)–($A4$) are sufficient to prove Theorem 10.1.12, asymptotic normality and efficiency of MLEs.

($A5$) For every $x \in \mathcal{X}$, the density $f(x|\theta)$ is three times differentiable with respect to θ, the third derivative is continuous in θ, and $\int f(x|\theta)\,dx$ can be differentiated three times under the integral sign.

($A6$) For any $\theta_0 \in \Omega$, there exists a positive number c and a function $M(x)$ (both of which may depend on θ_0) such that

$$\left|\frac{\partial^3}{\partial\theta^3}\log f(x|\theta)\right| \leq M(x) \quad \text{for all } x \in \mathcal{X}, \quad \theta_0 - c < \theta < \theta_0 + c,$$

with $\mathrm{E}_{\theta_0}[M(X)] < \infty$.

Consistency of the MLE

An estimator $\widehat{\theta}$ is consistent for θ if as the sample size n increases, the estimator converges to the true value of the parameter.

Formally, we would write: For any $\epsilon > 0$,

$$\lim_{n \to \infty} P(|\widehat{\theta} - \theta_0| > \epsilon) = 0$$

This is a fairly low standard placed on an estimator, and, fortunately, under certain "regularity conditions," the MLE is consistent.

Asymptotic Normality of the MLE

Another great property of the MLE: The MLE is asymptotically normal, with variance that is not difficult to calculate, under certain "regularity conditions." (See page 516 from Casella and Berger, Second Edition.)

This will allow an easy route to:

1. Calculate standard errors for MLEs
2. Constructing confidence intervals for parameters
3. Constructing confidence intervals for general functions of parameters (via the Delta method)
4. Constructing hypothesis tests for parameters

To start, we focus on the case where the parameter θ is one-dimensional.

The formal result can be stated as follows:

true value of parameter

If $X_1, X_2, \ldots, X_n$ are i.i.d. $f(x;\theta_0)$, then, under suitable regularity conditions, as n increases,

$$\sqrt{n}\,(\hat{\theta} - \theta_0) \longrightarrow N\big(0, I^{-1}(\theta_0)\big)$$

where

$$I(\theta_0) = E\bigg(-\frac{\partial^2 \log f(X;\theta)}{\partial \theta^2}\bigg)$$

is the Fisher Information.

Exercise: Explain the practical value of the above result.

$$\sqrt{n}\,(\hat{\theta} - \theta_0) \to N\big(0, I^{-1}(\theta_0)\big)$$

$$\sqrt{n}\,(\hat{\theta} - \theta_0) \approx N\big(0, I^{-1}(\theta_0)\big) \quad (\text{for "large" } n)$$

$$\hat{\theta} \approx N\Big(\theta_0, \frac{I^{-1}(\theta_0)}{n}\Big)$$

$$N\Big(\theta_0, 1/nI(\theta_0)\Big)$$

$E(\hat{\theta}) \approx \theta_0$, i.e. approx. unbiased. In fact, MLE is <u>asymptotically unbiased</u>, $E(\hat{\theta}) \to \theta_0$ as $n \uparrow \infty$.

$$V(\hat{\theta}) \approx 1/nI(\theta_0)$$

Exercise: Show that

$$nI(\theta) = E\left[-\frac{\partial^2}{\partial\theta^2}\log L(\theta)\right]$$

$$nI(\theta) = nE\left(-\frac{\partial^2 \log f(X;\theta)}{\partial\theta^2}\right)$$

$$= \sum_{i=1}^{n} E\left(-\frac{\partial^2 \log f(X_i;\theta)}{\partial\theta^2}\right) \quad \text{since } X_1, X_2, \ldots, X_n \text{ iid}$$

$$= E\left(-\sum_{i=1}^{n}\frac{\partial^2 \log f(X_i;\theta)}{\partial\theta^2}\right)$$

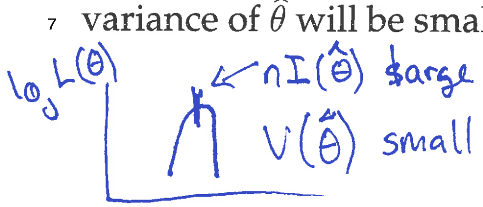
$$= E\left(-\frac{\partial^2 \sum_{i=1}^{n} \log f(X_i;\theta)}{\partial\theta^2}\right)$$

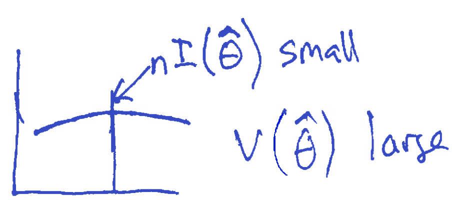
$$= E\left(-\frac{\partial^2 \log L(\theta)}{\partial\theta^2}\right)$$

The previous result shows that the variance of the MLE is approximately equal to the reciprocal of the expected value of the negative second derivative of the log likelihood at its peak.

And, "the negative of the second derivative" is the amount of curvature at the peak of the likelihood.

If the log likelihood has a "sharp" peak, then $nI(\theta)$ will be large, and the variance of $\hat{\theta}$ will be small.





Exercise: Assume $X_1, X_2, \ldots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, where σ^2 is known. We know that $\overline{X}$ is the MLE for μ. Derive the Fisher Information for μ, and use it to approximate the standard error and distribution of $\overline{X}$.

$$\ell(\mu) = \frac{-\sum(X_i-\mu)^2}{2\sigma^2} + \text{constants which do not depend on } \mu$$

$$\frac{\partial \ell(\mu)}{\partial \mu} = \frac{\sum(X_i-\mu)}{\sigma^2} = \frac{\sum X_i - n\mu}{\sigma^2}$$

$$\frac{\partial^2 \ell(\mu)}{\partial \mu^2} = -\frac{n}{\sigma^2}$$

$$nI(\mu) = E\left[-\frac{\partial^2 \ell(\mu)}{\partial \mu^2}\right] = E\left(-\left(-\frac{n}{\sigma^2}\right)\right) = \frac{n}{\sigma^2}$$

So, the asymptotic normality result says that $\overline{X}$ (the MLE for μ) is approx. normal with mean μ_0 and variance of

$$\frac{1}{nI(\mu)} = \sigma^2/n$$

We already knew that $\overline{X}$ is exactly normal with mean μ_0 and variance σ^2/n

Also note that $I(\mu) = 1/\sigma^2$

Exercise: Assume that $X_1, X_2, \ldots, X_n$ are i.i.d. Exponential(λ_0). Derive the Fisher Information for λ and the approximate distribution of its MLE. $\hat{\lambda}_{MLE} = 1/\bar{X}$

$$f(x;\lambda) = \lambda e^{-\lambda x}, \quad x > 0$$

$$\log f(x;\lambda) = \log \lambda - \lambda x$$

$$\frac{\partial \log f(x;\lambda)}{\partial \lambda} = \frac{1}{\lambda} - x$$

$$\frac{\partial^2 \log f(x;\lambda)}{\partial \lambda^2} = -\frac{1}{\lambda^2}$$

So, $I(\lambda) = E\left[-\frac{\partial^2 \log f(X;\lambda)}{\partial \lambda^2}\right] = E\left[1/\lambda^2\right] = 1/\lambda^2$

and $nI(\lambda) = n/\lambda^2$ and $V(\hat{\lambda}_{MLE}) \approx 1/nI(\lambda_0) = \lambda_0^2/n \approx \hat{\lambda}^2/n$

we know $\hat{\lambda} = 1/\bar{x}$, so $V(\hat{\lambda}_{MLE}) \approx 1/n\bar{X}^2$ and $SE(\hat{\lambda}_{MLE}) = \frac{1}{\sqrt{n}\,\bar{X}}$

Exercise: Assume that $X_1, X_2, \ldots, X_n$ are i.i.d. Poisson(λ_0). Derive the Fisher Information for λ and the approximate distribution of its MLE.

$$f(x;\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, \ldots$$

$$\log f(x;\lambda) = -\lambda + x\log\lambda - \log x!$$

$$\frac{\partial^2 \log f(x;\lambda)}{\partial \lambda^2} = -\frac{x}{\lambda^2}$$

$$I(\lambda) = E\left(-\frac{\partial^2}{\partial \lambda^2}\log f(x;\lambda)\right) = E\left[\frac{X}{\lambda^2}\right] = \frac{1}{\lambda}$$

So, the MLE of λ ($\bar{X}$) has approx. variance of $1/nI(\lambda_0) = \lambda_0/n \approx \hat{\lambda}/n = \bar{X}/n$